

Can We Use Statistical Mechanics and Generative Artificial Intelligence to Improve the Retrieval Mechanism in Retrieval-Augmented Generation While Reducing the Reliance on Large Context Windows for Local Compute?

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Abstract: Large legal documents challenge Large Language Models because they are too long, too specialised and too expensive to pass into a context window for every query. This project investigates if we can use statistical mechanics and Generative Artificial Intelligence to improve the retrieval mechanism in RAG while reducing the reliance on large context windows for local compute. A corpus of five credit agreements was analysed using Zipf's law, Heaps' law and Shannon entropy which revealed a repetitive legal vocabulary that proved challenging to create reliable training supervision from. The project therefore used Gemma3-12B to generate synthetic query–chunk pairs locally to fine-tune the embedding model E5-base-v2, using section-aware chunking with hybrid dense/BM25 hard negatives under five-fold leave-one-document-out cross-validation. Recall@10 increased from 0.637 to 0.797, Mean Reciprocal Rank went from 0.387 to 0.483 and the mean rank fell from 45.96 to 14.77. Relating the cosine similarity to negative energy results in a Boltzmann like retrieval distribution. At a $\tau = 0.05$ retrieval entropy dropped from 10.49 to 4.75 bits and the top-5 probability mass increased from 0.013 to 0.589 which resulted in a 68% decrease in required tokens at 80% recall accuracy. Statistical mechanics did not generate enough training signal alone, but it explained why generative supervision was needed and provided a quantitative insight into how fine-tuning changes the embedding landscape. These results show that thermodynamic diagnostics, when combined with local generative supervision, can substantially improve retrieval accuracy on local hardware without relying on expensive cloud-based context windows.

Introduction

Large Language Models (LLMs) are becoming more popular and are now regularly being applied across physics [1]. Neural networks are used to extract hydrodynamic parameters straight from active matter experiments [2] and machine learning models are fine-tuned to analyse molecular dynamics trajectories [3].

In all these situations, researchers encounter the same problem: the performance of an off-the-shelf model is insufficient and a model must be domain adapted. This project encounters the same challenge and applies the same problem solving to a legal setting.

Long-form legal documents test the current limits of automated retrieval. Credit agreements cover hundreds of pages with cross-referenced clauses, nested definitions and multi-party financial schedules, as a result, they cannot be summarised or broken up randomly as this will ruin the legal interpretation of the whole document. The natural first response, especially now, is to feed the whole document into an LLM as context. Models with large enough context windows exist, with frontier models accepting more than 1 million tokens, so this is possible, but their computational cost scales quadratically with the full

document length, which is proportional to $\mathcal{O}(N^2)$, where N is the document length [4]. Every follow up question means the whole document needs to be encoded again to find an answer as there is no working memory that persists over time. Other than additional inference cost, long-context models suffer from lost in the middle behaviour [5], where the retrieval accuracy for facts in the middle of the context window is degraded, often leading to hallucination where the model generates seemingly correct information which is actually false. This poses a critical problem for credit-agreements as most of their operative content is in the middle section of the documents. An industry credit-agreement retrieval pipeline that motivates this work achieves 60–70% retrieval accuracy using an off-the-shelf all-mpnet-base-v2 embedding model requiring a human analyst to verify results whenever retrieval fails.

Retrieval-Augmented Generation (RAG) was designed as an alternative to the purely LLM-based approach [6]. The approach is simple in principle, a retriever identifies the k passages most relevant to the user's query from the document and then only that information is passed into the LLM as context. After the initial embedding of the document, which

scales linearly $\mathcal{O}(N)$, the context size for following requests becomes $\mathcal{O}(kc)$, where c is the mean chunk length. In long credit agreements, where $N \gg kc$, retrieval quality is therefore of the utmost importance. If the correct passage is not part of the top k retrieved chunks, then the LLM will not have the specific knowledge needed to create an answer.

Standard RAG pipelines are built using general-purpose embedding models that are trained on everyday human text and broad web corpora. As a result, the latent geometry of the underlying vocabulary does not align with legal domain causing poor retrieval results out of the box.

This report addresses the following research question, posed by an industry partner:

Can we use statistical mechanics and Generative Artificial Intelligence to improve the retrieval mechanism in RAG while reducing the reliance on large context windows for local compute?

The question is answered by three subclaims. First, a local generative Artificial Intelligence (AI) model can create enough supervision signal to not need manual annotation, cloud Application Programming Interfaces (APIs) or have data leave our secured environment. Second, domain-specific fine-tuning of an embedding model using synthetic query–chunk pairs will substantially improve retrieval quality compared to off-the-shelf baselines. Third, a sharper retrieval distribution, with probability mass concentrated in fewer more relevant chunks, reduces the context window required downstream which addresses the computational half of the research question.

A statistical-mechanics perspective

Statistical mechanics will be applied across sections and will be a diagnostic tool for the observations made. The rank–frequency structure of the corpus can be determined by Zipf’s law with vocabulary growth by Heaps’ law and Shannon entropy describing why lexical supervision struggles in this domain. Legal text is filled with high-frequency words which create a latent space where semantically unrelated clauses can be clustered together. This overlap makes creating reliable positive–negative separation from statistics alone challenging and is what motivates explicit semantic supervision. After fine-tuning, the statistical view is reapplied where we relate the Multiple Negative Ranking Loss (MNRL) to a Boltzmann distribution over chunk similarities in which the cosine similarity is the negative energy

and the softmax scaling factor acting as the inverse temperature. Fine-tuning can therefore be interpreted as lowering the free energy of the correct query–chunk pairs and this can be measured through the resulting reduction in retrieval entropy.

The contributions of this work are:

1. A section-aware approach to chunking for long legal PDFs, which is built to preserve high-entropy legal sections, as identified through Shannon entropy analysis.
2. 12,520 synthetic query–chunk training triplets created locally by Gemma3-12B via Ollama without human annotation or external APIs
3. A detailed comparison of embedding models, chunking strategies, and negative-mining strategies under five-fold leave-one-document-out cross-validation, resulting in a 25.2% improvement in Recall@10.
4. A statistical mechanical interpretation of retrieval being an energy based probability distribution which links the reduction in retrieval entropy to the reduction in required context tokens.
5. A demonstration that a full pipeline is possible without enterprise infrastructure: synthetic triplet generation was done on an Apple M4 MacBook Air with 16 GB of unified memory, fine-tuning ran on a single university GPU (NVIDIA RTX 4000 Ada with 20 GB of VRAM) making the approach accessible for local and privacy-preserving deployment.

Background and Theory

Dense embeddings and cosine similarity

An embedding model, $f: \mathcal{X} \rightarrow \mathbb{R}^d$, maps a chunk of text, x , to a d -dimensional vector, $\mathbf{v} = f(x)$. For E5-base-v2 (E5), our chosen embedding model, this is a 768-dimensional space. ℓ_2 normalisation confines each vector to a unit sphere which lets the angle between two vectors fully determine their relationship allowing retrieval to be calculated using cosine similarity:

$$S(\mathbf{v}_i, \mathbf{v}_j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|}, \quad (1)$$

where $S = 1$ are vectors pointing in the same direction (semantically identical text) and $S = 0$ indicates orthogonal vectors (no shared meaning).

For retrieval to take place, the document chunks need to be embedded and then stored either in a flat file or a vector database depending on the scale.

When a user asks a question, the query is embedded into the same embedding space as all the chunks. This cosine similarity is calculated between the user query and each chunk, forming a pair, that is then ranked from most similar to least. Relevant query–chunk pairs should occupy the same area of embedding space, creating a cluster, while semantically unrelated chunks should be pushed apart [7, 8]. This baseline separation in embedding space is direct consequence of the type of data the model was trained on.

Quantitative linguistics of legal text

Zipf’s law [9], describes the rank–frequency distribution.

$$f(r) \propto r^{-\alpha}, \quad (2)$$

Heaps’ law [10] describes vocabulary growth:

$$V(N) = KN^\beta, \quad (3)$$

where β is the Heaps exponent. Shannon entropy [11], identifies information-dense regions.

$$H_i = - \sum_k p_{ik} \log_2 p_{ik}, \quad (4)$$

These measurements are useful and provide a statistical description of the legal lexical landscape of the documents, which helps inform the best chunking approach before we apply embedding-based retrieval. Together, these two things guide how the document should be split, ensuring that legal context is preserved.

Contrastive learning as a Boltzmann distribution

To realign the underlying geometry to better suit the legal domain, E5 is fine-tuned using contrastive learning. Contrastive learning is a process where you get the model to compare positive and negative pairs ensuring the correct chunk is pulled closer to the query, increasing its similarity, while all other negatives in the batch are pushed away, reducing their similarity. For a query q , a positive chunk p^+ , and a set of in-batch negatives $\{p_i^-\}_{i=1}^N$, the MNRL is [12, 13, 14]:

$$\mathcal{L} = -\log \frac{\exp(s S(q, p^+))}{\exp(s S(q, p^+)) + \sum_{i=1}^N \exp(s S(q, p_i^-))}, \quad (5)$$

where s is a softmax scale parameter. Minimising the loss, $\mathcal{L}$, is the process of increasing the similarity between the query and the correct chunk while simultaneously reducing the similarity between all other

negatives in the batch. The choice of MNRL over another loss, such as triplet loss, is intentional. They do the same thing, pull positive pairs together and push away negatives, but, MNRL pushes away all other negatives in the batch and these extra negatives increase the number of effective training signals per gradient step, making MNRL highly efficient.

This loss function has a direct thermodynamic interpretation. If we treat the effective energy for each candidate chunk as:

$$E_i = -S(q, p_i), \quad (6)$$

then the probability that chunk p_i is successfully retrieved for query q takes the shape of a Boltzmann distribution:

$$P(p_i | q) = \frac{\exp(-E_i/\tau)}{\sum_j \exp(-E_j/\tau)} = \frac{\exp(S(q, p_i)/\tau)}{Z}, \quad (7)$$

where $\tau = 1/s$ is the effective temperature parameter and $Z = \sum_j \exp(-E_j/\tau)$ is the partition function. At $s = 20.0$, the default value for MNRL, the effective temperature is $\tau = 0.05$.

The retrieval entropy,

$$S_{\text{ret}} = - \sum_i P(p_i | q) \log_2 P(p_i | q), \quad (8)$$

is a measure of how spread out the probability mass is across the candidate chunks. A high-entropy retrieval distribution means that many chunks will look like they answer the user query, confusing the retriever. Alternatively, for a low-entropy distribution, the model assigns most of the probability to a small number of candidates, increasing the chances of retrieval success.

If we consider the Helmholtz-like free energy,

$$F = U - \tau S_{\text{ret}}, \quad (9)$$

where $U = \langle E \rangle$ is the mean energy, this provides a quantifiable way to measure the change fine-tuning has on our embedding model. Fine-tuning should lower the free energy of the correct query–chunk pair relative to the other negatives, as the probability mass becomes more concentrated, lowering the effective context window as fewer chunks are needed.

BM25 and hybrid retrieval

Best Matching 25 (BM25) is a sparse retrieval method that ranks passages by comparing the actual words from the user query, rather than the meaning of those words, to the words in the document. BM25 is similar

to another method, Term Frequency–Inverse Document Frequency (TF–IDF), but it also includes a term for frequency saturation and a correction for document length [15], which allocates larger scores to chunks that contain important query words, especially when those words occur rarely across the collection of documents. The score increases when a query term appears in a chunk, but repeated terms contribute less after a certain point, preventing long or repetitive chunks from being over-rewarded.

For credit agreements, using BM25 alone has limitations. Important legal terms, such as “Borrower”, “Lender”, “Agreement”, and “Section”, appear in many different clauses across the document. A passage may rank highly because it shares word “Lender” with the user’s query, even if it addresses a different unrelated legal issue. Consequently, the method is hardly used as a sole retriever in industry and will not be in this project, but it still can provide useful training signal when paired with a dense retrieval method.

This hybrid approach does what is expected, combining the strength of key word matching from BM25 with the semantic understanding of dense retrieval. If the user asks a question about a specific case ID, dense retrieval is useless but BM25 matches it easily. If there is a question about “not being able to pay” but the agreement says “Events of Default”, BM25 fails, dense retrieval returns the correct result.

As these two methods rank their findings on two different scales, to combine the negatives we use the Reciprocal Rank Fusion (RRF) [16]. The RRF score for a document (d) is calculated as follows:

$$\text{RRF_score}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}(r, d)} \quad (10)$$

in this equation $\text{rank}(r, d)$ is the position of the document in list r and k is a constant (usually set to 60) that prevents top-ranked results from completely overwhelming the rest of the list.

Prior work in legal natural language processing

Prior work including LEDGAR [17], CUAD [18], ContractNLI [19] and ACORD [20] all agree that contracts are a difficult domain for standard Natural Language Processing (NLP) techniques and domain adaption is required for success. These data sets focussed on classification, contract review and clause retrieval, but they do not address improving local RAG retrieval for credit agreements through the use of synthetic supervision and statistical-mechanical analysis. This project

sits on the boundary between NLP and statistical physics.

Methods

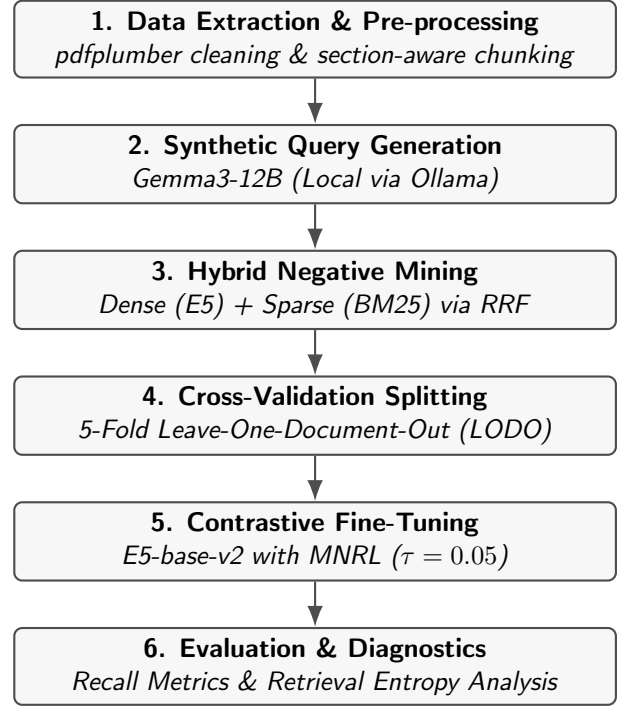


Figure 1: End-to-end local pipeline. Raw credit agreements are processed and split into section chunks, which then take part in synthetic query generation followed by hybrid negative mining. These triplets are split across five folds for contrastive fine-tuning, ending at retrieval and thermodynamic evaluation.

Corpus and preprocessing

Five sample credit agreements were provided by an industry partner to fine-tune the embedding model. Text was extracted from the PDFs using PDF Plumber and cleaned using regular expressions and fixed-text rules. After pre-processing, including the removal of OCR artefacts, page numbers and excessive white spaces, 448,331 cleaned tokens remained across the five agreements. A common NLP technique is the removal of stop words, which are commonly used, high-frequency words such as “the,” “is,” “and,” “in,” and “a,” that add little semantic meaning and can distort the underlying lexical landscape. In regular, everyday text, stop words make up 40–50% of a passage’s content [21]. Consequently, stop-word filtering and lemmatisation were applied at the corpus level, to ensure analysis with Zipf’s law, Heaps’ law, Shannon entropy, and n-gram frequency measurements aligned with the context of interest. After lemmatisation and stop-word

filtering, 244,557 tokens remained with 4,824 being unique tokens, a reduction of 45%. Despite this reduction in noise, it was a deliberate decision to keep the stop words for triplet generation, negative mining and fine-tuning as E5 was pre-trained on natural text that included them. With their removal, it risked a shift in the underlying latent geometry, compromising retrieval performance. See appendix for full stop-word breakdown.

Initial statistical route

The project first attempted to use a purely statistical approach, based on observations from pre-processing, to create positive and negative training pairs. The methods attempted included: local co-occurrence, bigrams (two-word sequences), trigrams (three-word sequences), nearest neighbour proximity and raw embedding cosine similarity. This approach did not bear fruit. After analysing the failure, the conclusion was that the highly repetitive style of credit agreements, with the same terms appearing throughout the documents carrying slightly different legal meanings, would make a purely statistical approach difficult to justify. The ambiguity motivated the change to generative supervision with the initial statistical lens being reapplied after fine-tuning to measure the change.

Chunking strategies

Chunking is a process where long passages of text are broken into smaller chunks, hence the name. There are many different methods of chunking, but their overall goal is the same: turn the data into a searchable library of self contained ideas. Two strategies were compared: paragraph chunking and section-aware chunking.

Paragraph chunking splits the text on paragraph boundaries, which appear as "`\n\n`" after pre-processing with `pdfplumber`. The approach yielded 2,137 chunks and was simple to implement but fragmented legal meaning and lost context over page breaks. A key oversight at the time was not including a token length cap, which caused many chunks to exceed the embedding model's token limit, resulting in truncation.

Section-aware chunking, used regular expressions to detect section headings in the credit agreements. It was designed to follow a five-stage procedure: (1) split at section boundaries, (2) detect and preserve definition blocks, (3) merge short neighbouring paragraphs within the same section, (4) split oversized sections at sentence boundaries, and (5) apply a hard

512-token limit. The approach produced fewer chunks, at 1,582, but the improvement in chunk quality meant that the chunks were now semantically coherent and will produce better training signal. We can relate section-aware chunking to a coarse graining operation, as it groups together low-level text units into larger macrostates, in this case, legal sections. The preservation of high-entropy regions, as identified by Shannon entropy, was a key motivator to proceed with this method.

Synthetic triplet generation

Gemma3-12B is a 12-billion-parameter instruction-tuned model run locally via Ollama. It is an open-weight LLM that does not require internet access, share data or need any API once downloaded. To create synthetic queries we first need a tailored prompt that would take in a chunk, understand its context and return an answer. The field of prompt engineering is not the focus of the project but its fundamentals play a key part. If the LLM produces poor synthetic queries, training signal are weak leading to poor retrieval. Consequently, the decision was made to use a frontier LLM to create the perfect prompt for our local LLM. Using GPT-5.4 by OpenAI, passing in relevant project context, desired outcomes and my best attempt at a prompt, a prompt fine-tuned to our case was created. The refined prompt included the section heading as additional context, as well as instructing the local LLM to create four distinct queries per chunk: precise factual, paraphrased, keyword-style search and scenario-based. Each was designed to mimic the various ways a human would ask questions. Each query-positive pair, along with its metadata, was cached for reuse. Full prompt included in appendix.

The final section-aware hybrid dataset contained 12,520 training triplets. This was produced from 1,565 section-aware chunks (17 short chunks removed), with four synthetic queries generated per chunk, giving 6,260 query-positive pairs. For each query-positive pair, two hard negatives were mined, one using dense retrieval and one using BM25 retrieval, resulting in $6,260 \times 2 = 12,520$ triplets.

Negative mining

Hard negatives, to make the training triplets, were mined separately for the section-aware pairs and the paragraph pairs. The baseline E5 model was used to embed the query-chunk pairs which meant that the mined negatives reflected the weaknesses of the original embedding space. For each pair, dense negatives

were mined at random, from ranks 3–20 for section chunks and 2–15 for paragraph chunks. In both cases, the highest ranked dense negatives were excluded intentionally as they could contain false negatives. False negatives confuse the contrastive learning process as they answer the query but they did not form part of the original query–chunk pair.

A second set of negatives were mined using BM25 for the section-aware pairs only, with candidates sampled from ranks 1–10 and combined with the dense negatives using RRF. The BM25 window did not omit any results as these negatives provide the model with contrastive examples of negatives with the same wording but different meaning. The combined hybrid mined negatives forced the embedding model to separate passages that are lexically similar but legally different. This approach directly targets the type of error the retriever needs to overcome and supports the project’s aim of improving retrieval for long legal documents.

Fine-tuning

E5-base-v2 (primary, 512-token limit) and GTE-ModernBERT-base (auxiliary, 8192-token limit) were chosen from the list of experimented models (Fig. 7, and fine-tuned using MNRL through the sentence-transformers library [7]. Training was implemented using the PyTorch’s machine-learning framework, with CUDA acceleration allowing tensor operations to run on the NVIDIA GPU. The training pipeline ran completely on accessible university hardware: an AMD Ryzen 9 7950X3D processor, 64 GB RAM and an NVIDIA RTX 4000 Ada GPU with 20 GB VRAM. The setup could run a max batch size of 16, so each query–positive pair would be compared against 15 other in batch negatives. The main hyperparameters are listed in Table 1.

E5 trained stably with a learning rate of 2×10^{-5} , reaching its best validation Recall@10 after two to three epochs across all five folds before early stopping. GTE required a lower learning rate of 5×10^{-6} to reduce training instability. A full five-fold evaluation was performed for E5 after fine-tuning, but due to time constraints, GTE was evaluated on only one fold. GTE is therefore treated as an auxiliary model rather than a direct five-fold comparison.

Evaluation protocol

E5 was evaluated using leave-one-document-out (LODO) cross-validation across 5 folds [22]. With only five agreements available, LODO allowed us to

Table 1: Training and evaluation configuration for the fine-tuned embedding models.

Hyperparameter	E5	GTE
Max tokens	512	8192
Learning rate	2×10^{-5}	5×10^{-6}
Batch size	16	16
Loss function	MNRL	MNRL
MNRL scale s	20.0	20.0
Max epochs	100	100
Early-stop patience	3	3
Evaluation folds	5	1

make more use of the data as it gave us 5 different training sets. In each of the 5 folds, one of the credit agreements was left out for testing, while the remaining 4 were used for training and validation. To ensure no data leakage, any triplet whose negative chunk came from the test document was removed before training. This is especially important because it prevents the model from seeing either positive or negative examples from the held-out agreement during training, reducing the risk of inflated evaluation scores.

The key performance metrics were Recall@10, Recall@1, Mean Reciprocal Rank (MRR) and mean rank. Recall@10 is the percentage of queries where the correct chunk appears in the top 10 retrieved results. Similarly Recall@1 measures the percentage of queries for which the correct chunk is ranked first. MRR is a measure of how early in the ranked list the correct chunk appears by averaging the reciprocal rank and finally, mean rank measures the average retrieval position of the correct chunk across all queries.

Thermodynamic diagnostics

After evaluation, the Boltzmann framework Eqs. (7)–(9) is applied to all the test queries, so we can measure the effect of fine-tuning on the underlying retrieval distribution. For each query q , the cosine similarity was calculated across 1565 section chunks and then converted into Boltzmann probabilities at an effective temperature of $\tau = 0.05$ ($s = 20$). Using this distribution, we calculated the retrieval entropy S_{ret} and the cumulative top- k mass M_k . To ensure that our result is not unique to the default MNRL operating point, τ is swept 70 log-spaced values from 0.01 to 2.0 measuring entropy and top- k mass at every value (Fig. 8). To ensure confidence in the result, queries were grouped by retrieval outcome: success ($r^+ \leq 10$) and failure ($r^+ > 10$), with their entropy distributions being compared to determine if retrieval entropy acts

as a per-query confidence signal. After this, if we consider the number of chunks k needed to reach a recall target, which is identified per fold and multiplied by the mean chunk length ($\bar{c} \approx 280$), we can convert these retrieval improvements into a context window reduction.

The retrieval system is not in thermal equilibrium, and the Boltzmann interpretation is applied as an analogy. $\tau = 1/s = 0.05$ is fixed by the MNRL scale parameter rather than any physical temperature.

Results

Corpus statistics

Figure 2 shows the statistics from the five agreements in the corpus. The Zipf plot shows a strong rank–frequency relationship, with $\alpha = 0.995$ for the top-500 tokens and $\alpha = 0.805$ after lemmatisation and stop-word filtering, with $R^2 = 0.996$ and $R^2 = 0.975$ respectively. In the raw corpus, the top 10 most common tokens account for 30.4% of all the tokens, with “the” alone being 6.4%. After removing stop words, the steep side of the rank frequency curve lowers, reducing the exponent to $\alpha = 0.805$. The top 10 share drops to 13.3%, now led by domain terms like “Lender” at 1.85% of filtered tokens. This flattening shows that the remaining legal vocabulary sits in a compressed mid-frequency band with other common terms being: “Lender”, “Borrower” and “Default”.

The Heaps’ law plot gives $V(N) = 84.70N^{0.327}$ with $R^2 = 0.984$. The exponent $\beta = 0.327$ shows that the number of new words being introduced grows relatively slowly as the number of tokens in the corpus increases, in comparison to typical English text, which sits between $\beta = 0.4$ to 0.6 . β can therefore be thought of as the creativity of a document. Understanding the structure of a credit agreement explains this Heaps’ law exponent. Credit agreements usually start with introductory material and definitions, which introduce a lot of new terms quickly that are then reused throughout. A lower β can also be seen as the difficulty of retrieval, as it indicates that the same words appear in many places, making accurate retrieval challenging.

The entropy figure provides the second part of this explanation, where an entropy distribution is observed with a mean of 6.015 bits per word, a median of 6.208, a standard deviation of 0.735 and a range from 0 to 7.225 bits per word across 1,582 section chunks. The low-entropy regions correspond to what is called boilerplate text, while high-entropy sections represent

dense definitions, financial details or schedules.

The entropy profile across normalised document position Figure 3, shows that the high entropy content is distributed throughout the document instead of being concentrated at either end. This observation is consistent with retrieval challenges caused by lost-in-the-middle behaviour on large context models.

The switch from paragraph chunking to section-aware chunking was made to ensure that these high-entropy regions were not split too aggressively, because a missing phrase, in a high-entropy region, can change the meaning of the whole section.

Why statistical supervision failed

The original aim of the project was to use corpus statistics to create training data automatically. If proximity and bi/tri-gram overlap had been able to reliably separate related pairs from unrelated pairs, the project would most likely have stayed with the statistical approach. However, the results at the time did not support this. In the Zipf plot, the high-frequency terms in the top left are not stop words. They are legally important, but they occur and reoccur everywhere, which makes it difficult to find relevance across the corpus. Despite this negative result, it is still an important result because it stops the physics in this report from being purely decorative and actually uses it to diagnose why a standard NLP approach failed. The legal corpus is not random, but it is also not organised in a favourable way where frequency statistics alone can identify semantic question–answer relationships.

Retrieval performance after fine-tuning

The main retrieval result is shown in Fig. 4 and Table 2. For section-aware triplets, fine-tuning improved E5’s Recall@10 from 0.637 to 0.797 on average across five held-out folds. This is a relative gain of 25.2%. Recall@1 improved from 0.265 to 0.331, MRR improved from 0.387 to 0.483 and mean rank fell from 45.96 to 14.77. The drop in mean rank is especially important because it shows that fine-tuning improved even the difficult cases where the correct chunk was far down the ranked list. Fig. 5 shows the consistent improvement across all five folds for Recall@10 and MRR. As only five documents were provided, it is crucial that the improvements all align, because even if two fold regressed for example, it would mean that 40% of the time, fine-tuning did not work. However, in every case, the model improved after fine-tuning, which suggests that it is more domain-aware. This is a core result for the project.

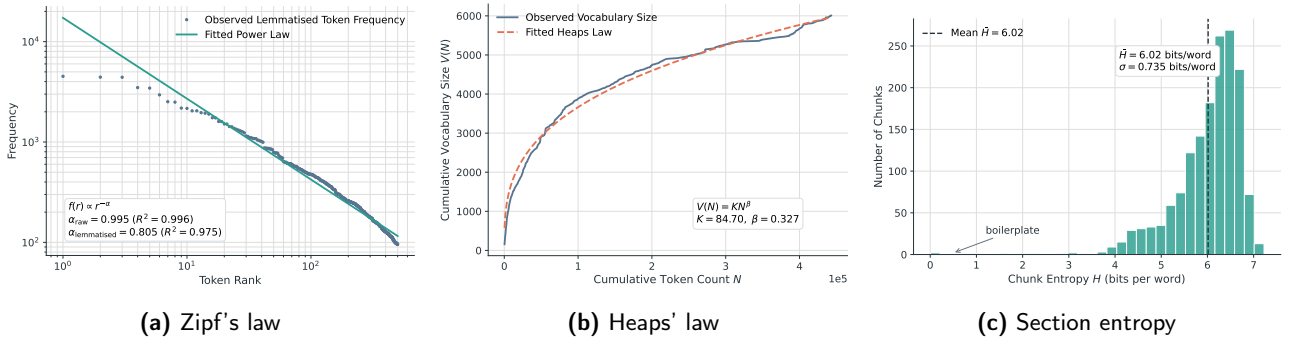


Figure 2: Corpus level statistical properties of the credit agreement dataset.

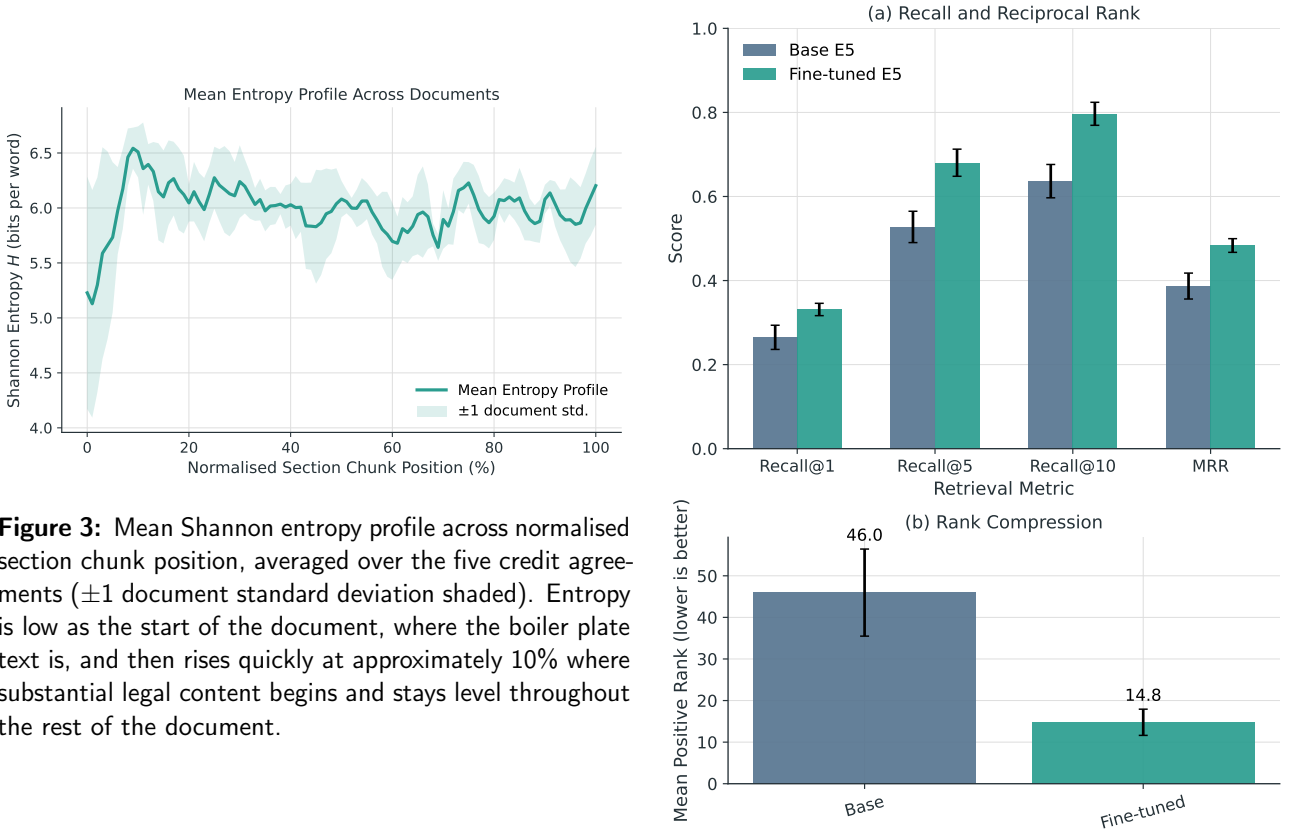


Figure 3: Mean Shannon entropy profile across normalised section chunk position, averaged over the five credit agreements (± 1 document standard deviation shaded). Entropy is low as the start of the document, where the boiler plate text is, and then rises quickly at approximately 10% where substantial legal content begins and stays level throughout the rest of the document.

Table 2: Five-fold retrieval performance: section-aware chunking, hybrid negatives. Values are cross-fold mean $\pm$ standard deviation.

Metric	Base E5	Fine-tuned E5	Δ
Recall@10	0.637 ± 0.040	0.797 ± 0.027	+0.160
Recall@5	0.528 ± 0.037	0.680 ± 0.032	+0.152
Recall@1	0.265 ± 0.029	0.331 ± 0.015	+0.066
MRR	0.387 ± 0.031	0.483 ± 0.016	+0.096
Mean rank	45.96 ± 10.47	14.77 ± 3.15	-31.19

Figure 4: Five-fold mean retrieval metrics for E5 before and after fine-tuning on the section-aware hybrid dataset. Error bars show one standard deviation across all folds.

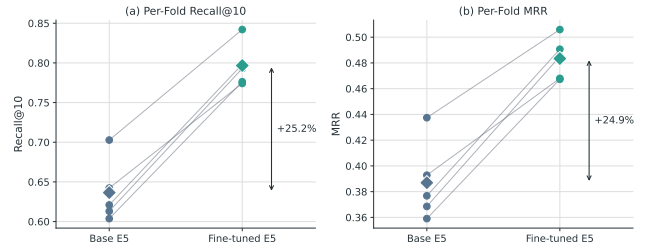


Figure 5: Fine-tuning improves Recall@10 and MRR in every one of the five held out folds. Each grey line connects the fold's base score to its fine-tuned score and the diamond marker shows the mean across all folds.

Embedding landscape before and after fine-tuning

Retrieval margin is the difference between the cosine similarity of the correct chunk, S^+ , which is the query-positive pair, and the hardest mined negative, S^- , defined as the negative with cosine similarity closest to 1:

$$\Delta S = S^+ - S^- \quad (11)$$

Figure 6 shows that fine-tuning increased the mean margin from 0.018 to 0.187 and the median margin from 0.016 to 0.174. This increase in similarity separation shows that, before fine-tuning, the correct chunk and hard-negative distributions overlapped heavily, while after fine-tuning, they separated substantially. This is the geometric reason why fine-tuning improves retrieval accuracy: the correct chunk becomes more separated from the plausible but incorrect alternatives. Figure 6 also gives another point of view as to why the statistical route failed. Before fine-tuning, the embedding model had grouped many legal chunks together because they shared the same wording and structure. However, after fine-tuning, when the model had seen synthetic examples of how a user might ask a question, the geometry of the embedding landscape widened, as the model was able to pick up on the nuances of the legal text.

However, the embedding landscape did not only change in a positive direction. In some cases, it changed in a negative direction. Figure 7 shows a query-rank improvement waterfall, where positive values indicate that the fine-tuned model ranked the correct chunk higher than the base model:

$$\Delta r = r_{\text{base}} - r_{\text{fine-tuned}} \quad (12)$$

The use of a symmetric log axis was chosen to ensure that even the smaller movements are visible and do not get visually crushed. The queries are sorted from the biggest improvement to the biggest worsening. Table 3 summarises the rank-improvement statistics.

Table 3: Summary of query-rank changes after fine-tuning.

Metric	Value
Total queries	12,520
Improved	6,856
Unchanged	3,052
Worsened	2,612
Median rank improvement	+1
Mean rank improvement	+30.7
Largest improvement	+1428 ranks
Largest worsening	-571 ranks

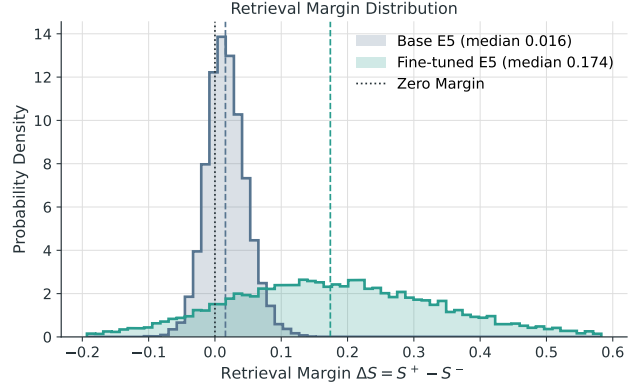


Figure 6: Fine-tuning shift the E5 retrieval margin distribution to the right, increasing separation between the correct chunk and the mined hard negative. The retrieval margin is calculated as $\Delta S = S^+ - S^-$.

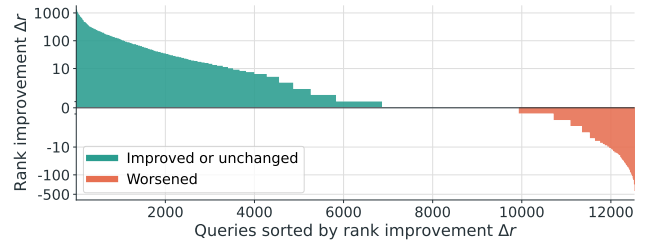


Figure 7: Each bar is a single query, sorted by $\Delta r = r_{\text{base}} - r_{\text{fine}}$. Positive values show that fine-tuning moves the correct chunk upward in the ranking, negative values indicate worsening. Symlog scale shows the small gains and rare large movements without hiding failures.

Failure case analysis

A failure was categorised as a situation where the correct chunk was still outside the top 10 even after fine-tuning. From this set, the 80 most severe failures were chosen and inspected then sorted first by the fine-tuned rank of the correct chunk, in descending order, and then by retrieval margin in ascending order. With this setup, it prioritised cases where the correct chunk was still furthest down the ranked list and where the model showed the weakest separation between the correct chunk and the mind hard negative.

Each of the 80 cases were assigned a failure mode which considered definition-like language, section length, lexical overlap between the positive and hard-negative chunks, lexical overlap between the query and positive chunk, retrieval margin and the presence of repeated boilerplate terms. These labels are generic and are meant for qualitative analysis only.

Table 4: Failure mode breakdown for the 80 worst fine-tuned E5 failures. Categories are diagnostic labels.

Failure Mode	Count	%
Hard negative genuinely similar	31	38.8
Definition ambiguity	26	32.5
Repeated boilerplate	14	17.5
Section-boundary issue	6	7.5
Weak synthetic query	3	3.8
Total	80	100.0

Auxiliary GTE-ModernBERT comparison

GTE-ModernBERT-base was included as an auxiliary comparison because it performed well as a base retriever during the initial search for an improved embedding model. However, due to time constraints and extra computational difficulty, a full five-fold fine-tuned evaluation was not possible for GTE. Table 5 presents the comparison between E5 and GTE across fold-one fine-tuning results only. Recall@10 went from 0.621 to 0.798 for E5 and from 0.659 to 0.767 for GTE. The fine-tuned GTE model actually achieved slightly higher Recall@1 and MRR for fold one, but lower Recall@10 and higher mean rank than fine-tuned E5. It was more difficult to work with GTE and it required a lower learning rate. However, the comparison between these two models should not be over interpreted. The reason for keeping E5 as the primary model was that it better aligned with the industry partner’s current pipeline, and if any integration were to happen, a fine-tuned E5 model would be able to

replace their existing embedding model more easily, as well as being able to run comfortably on local compute. GTE may still be worth further investigation, especially since it has a larger context window, which should be beneficial for the sections that did not fit into the 512-token limit of E5, as per the failure case.

Table 5: Fold-1 comparison: E5 vs GTE (section-aware, hybrid negatives). GTE evaluated on fold 1 only; results are indicative.

Metric	E5 Base	E5 FT	GTE Base	GTE FT
Recall@10	0.621	0.798	0.659	0.767
Recall@1	0.252	0.324	0.276	0.340
MRR	0.377	0.480	0.399	0.482
Mean rank	49.71	15.23	44.91	21.18

Thermodynamic retrieval diagnostics

If we apply the Boltzmann interpretation that we set up earlier, Eqs. (6)–(9), to E5 before and after fine-tuning, we can see how the underlying distributions have shifted. Using this we are able to compare our result to the context window problem that was mentioned in the research question.

Figure 8 shows how the mean retrieval entropy and the cumulative probability mass change as the effective temperature is swept. The MNRL has a default value of $\tau = 0.05$ and this is marked by a vertical dashed line.

In the retrieval disorder plot on the left, at $\tau = 0.05$, the base model has an average retrieval entropy of 10.49 bits, while the fine-tuned average is 4.75 bits, a reduction of 5.74 bits. Focusing on the right, the cumulative probability mass in the top 5 chunks, at the same value of τ , increase from 0.013 to 0.589.

A way to visualise this change in entropy is by viewing the change in cosine similarity of the query–positive and query–negative pairs before and after fine-tuning. In Figure 9, before fine-tuning, both (q, p^+) and (q, p^-) are very far to the right hand side of the similarity spectrum, with both peaking at 0.832 and 0.816 respectively. An important point to note is that the probability density of S^- is actually larger than S^+ . This is not a mistake and it is due to how we define S^- as being the chunk whose similarity is closest to 1, excluding our positive. As a result every term in the partition function,

$$Z(q) = \sum_j \exp(S_j/\tau), \quad (13)$$

contributes roughly the same amount and even

with the strong amplification of $s = 1/\tau = 20$, many chunks retain non-negligible Boltzmann weight and this keeps entropy high. This systematic shift explains the weakness behind the baseline E5 model when applied to an unfamiliar domain. Before fine-tuning, each chunk is clustered in a small region of embedding space as E5 was training on general web text and never learned to separate similar legal clauses.

However, after fine-tuning, the model shifts both distributions to the left and flattens them. The key observation now is that similarity for S^+ is further to the right than S^- with their medians being 0.433 and 0.249. This shift is exactly what caused the larger retrieval margin shown in Figure 6. Consequently, the resulting margin grows from a median of 0.016 to 0.174. At $\tau = 0.05$, the margin of 0.174 is amplified by the exponential factor to give a probability ratio of $\exp(0.174/0.05) \approx 32$, which means that the correct chunk contributes roughly 32 times more strongly to the partition function than the hardest negative and this causes entropy to collapse.

Discussion

What the project showed

The project started off by asking whether statistical mechanics could be used to understand the underlying structure of legal corpora in a way that could generate reliable positive and negative training pairs. It did reveal a structure, but not one that took the shape originally hoped. Zipf’s law, Heaps’ law and entropy showed that credit agreements have measurable statistical regularities, but the corpus was too lexically repetitive and too semantically confounded for the purely statistical approach. To reiterate, this project is not a failure from a physics perspective; it is a result produced from the physics perspective. The detailed statistical analysis diagnosed the problem and showed why lexical methods and proximity-based pair generation were likely to struggle. It then moved the project into the more emerging field of generative AI for synthetic supervision. After fine-tuning, the same statistical-mechanical lens was reapplied to examine how the underlying landscape had changed. This gives the project a coherent arc: statistical mechanics diagnoses the problem, generative AI supplies supervision, and statistical mechanics interprets the improved retrieval distribution.

Why generative supervision helped

Generative AI helped build the semantic bridge between a potential user query and the corpus chunks. It allowed the project to scale to thousands of queries. A process that would have taken a human days of work can now be done in an hour on a MacBook Air with no dedicated GPU. A section in a credit agreement is not just a block of text; it is evidence that can answer a range of possible questions. However, the raw corpus does not contain these questions explicitly. Gemma3-12B makes queries based on styles that were specifically designed to mimic the types of questions a user might ask when trying to retrieve information. This is very important in the legal domain, where exact wording varies and the reporting requirement, obligation, threshold and definition may all point to the same section.

However, there is still a limitation with this approach. The synthetic queries cover four possible styles of user questions, but this does not account for all the different types of question a user may ask. As a result, queries written by legal analysts may be more suitable for training the model in these rare instances and a human-in-the-loop system that provides these obscure or rarer questions would likely benefit the overall system. Regardless, the improvement over five-folds is enough evidence that the use of generative AI can provide a useful training signal and it should be considered a strong proof of concept.

Why section-aware chunking matters

The pivot to section-aware chunking was not just a formatting choice. The original method of paragraph chunking had a critical flaw: it did not implement any max token limit of 512. As a result, a substantial portion of each chunk was truncated and its meaning not fully captured. Section-aware chunking fixes this and includes a token cap, preserves legal meaning by splitting at section boundaries and provides the section heading as additional context.

However a direct comparison between the two methods would not be fair as they differ across three key variables simultaneously: chunking strategy, corpus size and negative mining method. As a result the improvement in retrieval cannot be solely attributed to chunking alone without being able to perform some sort of ablation. Despite this, the qualitative argument for section aware chunking is strong: it preserves the high entropy legal structure and it does not truncate any sections.

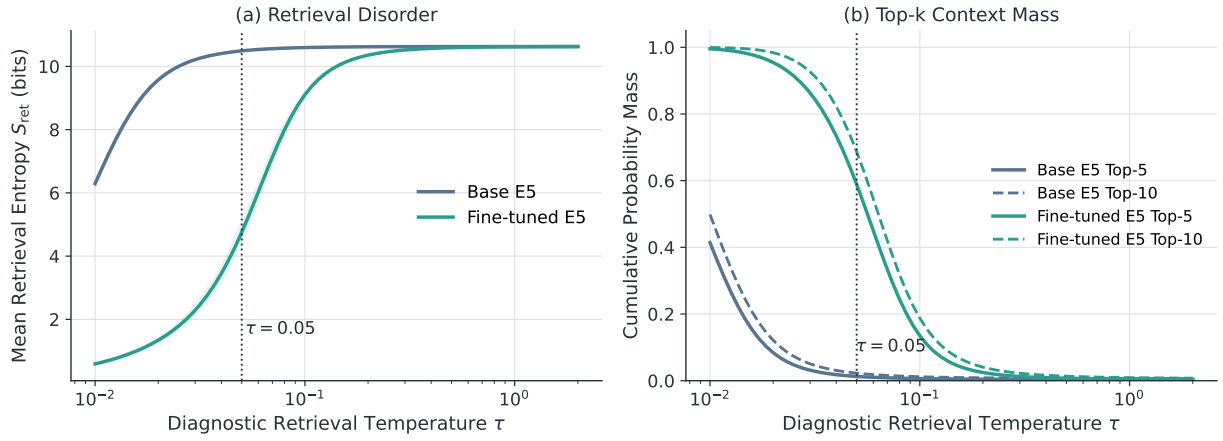


Figure 8: Boltzmann diagnostic quantities swept over inverse temperature τ . The dotted vertical line marks $\tau = 0.05$, corresponding to MNRL scale $s = 20.0$. Fine-tuned E5 shows lower retrieval entropy and higher top- k probability mass across all temperatures.

Comparison with prior literature

The results presented here align with the ranges reported in prior dense retrieval and legal NLP. Karpukhin et al. [8] showed that fine-tuning a dense retriever with specific question–passage pairs can improve Recall@20 scores by up to 19% over BM25 on Natural Questions. The +16.0% gain seen on our Recall@10 is consistent with this scale of improvement and is achieved on only five documents rather than a large annotated dataset like in [8]. CUAD[18] shows that contract understanding remains challenging for Transformer models and performance is heavily affected by model design and training data. In ACORD[20], it is shown that contract clause retrieval is still difficult and notes that a way to improve is through domain specific retrieval.

The size of the improvement made aligns with recent financial-domain embedding adaptation work. Brenner et al. [23] report a 27.7% relative improvement in MRR@5 from LLM-distilled fine-tuning on SEC filings. In this project, fine-tuning E5-base-v2 on synthetic credit-agreement triplets gives a comparable 27.3% relative improvement in MRR@5 (derived) and a 25.2% relative improvement in Recall@10. However we cannot make a clear comparison as Brenner et al. train at a much larger scale with a much more capable embedding model whereas this work used five agreements and locally generated synthetic supervision. Nevertheless, the similar relative gain between our work indicated that LLM-generated supervision can effectively adapt off the shelf embedding models to legal credit agreements without manually annotating datasets.

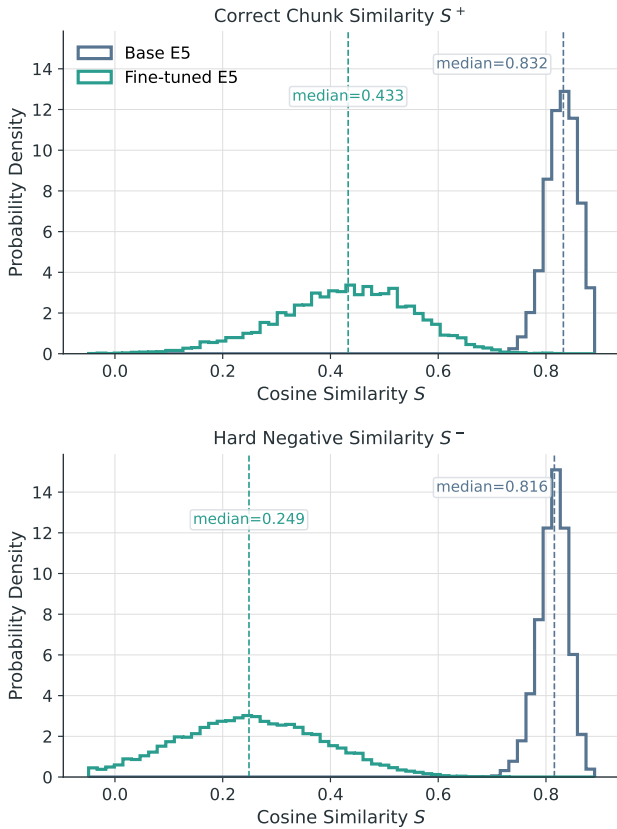


Figure 9: Cosine similarity distribution for correct chunks (S^+ , top) and hard negatives (S^- , bottom) before and after fine-tuning. Before, both distributions peak at around 0.82, giving near identical similarities to correct and incorrect chunks. After the fine tuning, S^+ drops to a median of 0.433 and S^- drops to 0.249 which creates the margin shown in Figure 6.

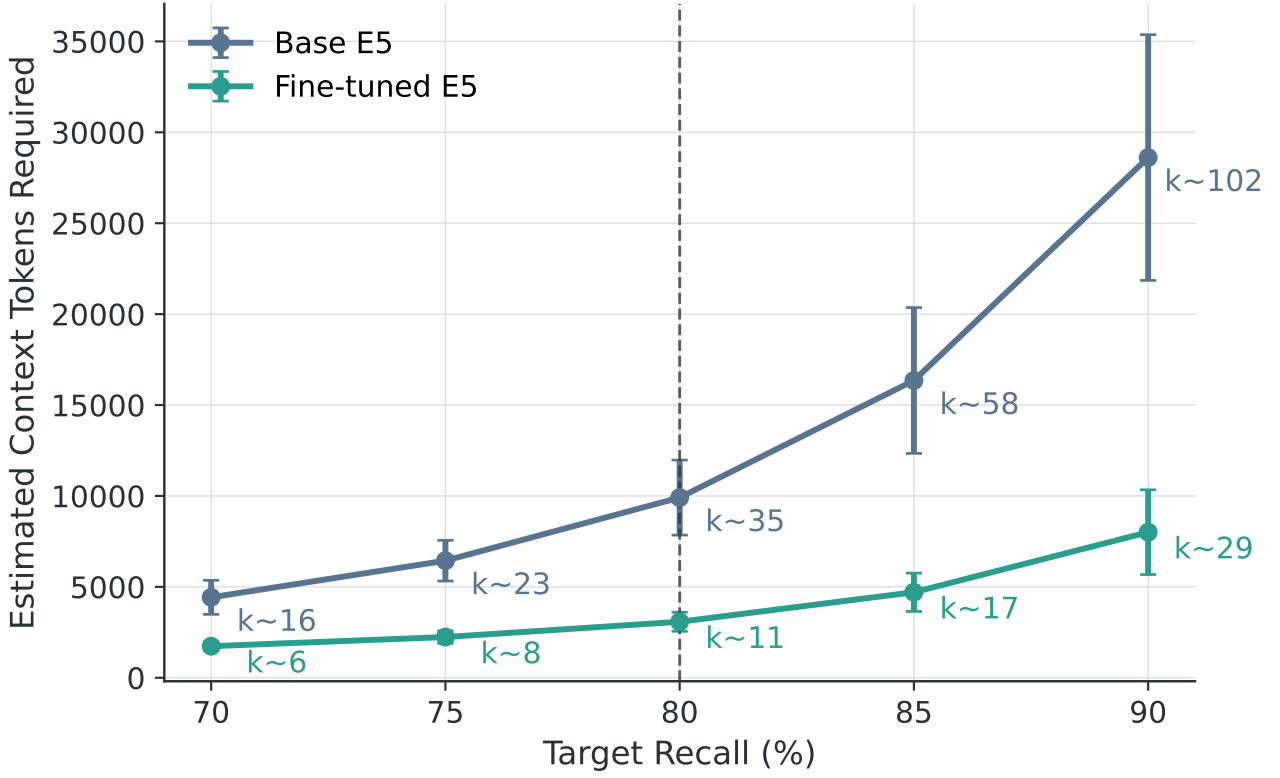


Figure 10: Estimated context tokens ($k \times \bar{c}$, mean chunk length $\bar{c} \approx 280$ tokens) needed to reach each target Recall@ k . Fine-tuned E5 achieves 80% recall with $k \approx 11$ chunks ($\approx 3,080$ tokens) with the base model needing $k \approx 35$ ($\approx 9,800$ tokens).

Physics interpretation

The physics interpretation is best applied to the retrieval entropy distributions before and after fine-tuning. Before fine-tuning, the model has a high-entropy distribution, where many chunks look similarly plausible and the probability mass is spread wide (as evidenced by the large overlap at high similarities, Figure 9). After fine-tuning, this changes, and the correct chunk shifts towards a lower effective energy causing the retrieval margin to increase (Figure 6) and the distribution to sharpen. In Figure 8, at $\tau = 0.05$, retrieval entropy drops from 10.49 to 4.75 bits, while the top-five probability mass rises from 0.013 to 0.589.

This relates directly to the context window problem. A sharper retrieval distribution means that fewer chunks need to be passed into the LLM to reach the same recall target as shown in Figure 10. This figure explains that the number of chunks (k), needed to achieve a target recall (R), can be found by using the smallest rank cut-off that is needed to ensure that the correct chunk is retrieved $R\%$ of the time. So, to explain it with an example: if we test the system using 100 different queries, we want the correct answer to be retrieved for 80 of them. For the base model, the correct chunk was ranked 35th or better 80% of the

time. This means that our LLM, when we want to get an answer from it, will need to retrieve the top 35 chunks (9,800 tokens) to hit the 80% target. After fine-tuning, the correct chunk is now ranked 11th or better 80% of the time and our LLM now will only need to retrieve the top 11 chunks (3,080 tokens). This results in a 68% context window reduction.

To understand this 68% context window reduction, we can look at the model's improved efficiency through a statistical mechanics view. The effective temperature $\tau = 1/s = 0.05$ is fixed to the default MNRL scale parameter and this allows us to apply a Boltzmann distribution as a mapping. Using this relationship, we are able to equate the cosine similarity to the negative energy, the softmax denominator acts as our partition function and the retrieval entropy represents the thermodynamic disorder. By looking at fine-tuning through this lens we can see that it lowers the effective energy of the query-chunk pairs, causing probability mass to concentrate, which drops the entropy, allowing us to measure the effect of fine-tuning by calculating the change in retrieval entropy (ΔS_{ret}).

Limitations

The first limitation would be the size of the corpus. Only five credit agreements were available, producing 12,520 training triplets. This number of triplets is relatively small compared to the size of the embedding model but two design choices were made to mitigate this: LODO cross validation ensured that every document was used for both training and testing; and MNRL’s ability to learn from all other 15 negatives in the batch simultaneously multiplied the useful training signal per gradient step. The power of MNRL was limited by the GPU’s VRAM (20 GB) with a batch size of 16 using 17 GB. Despite this, a transformer of E5’s size would typically see many more training examples, and the relatively low Recall@1 improvement from 26.5% to 33.1%, compared to the Recall@10 gain, suggests that the model is not yet confident enough. More documents would probably close the gap.

The second limitation is the relatively low complexity and limited coverage of the synthetic queries. This step made the project feasible and allowed it to scale without manual annotation, but the queries might be too clean compared to real analyst questions. Real analyst questions may include abbreviations, partial clause references, or domain shorthand, and the four styles used here, factual, paraphrased, keyword, and analytical, do not cover exactly how an analyst would query a credit agreement. A next step would be to evaluate the model on a small set of expert written queries and test if similar improvements are noticed.

The third limitation is that GTE was only fine-tuned and evaluated on one fold. This made it difficult to compare directly with E5, hence its classification as an auxiliary comparison.

Future directions in domain-specific retrieval

Being able to successfully generate local generative supervision allows for promising future developments in the legal NLP space. One of the main ways forward is to move from the standard Transformer-based bi-encoders to sub-quadratic architectures. The current bi-encoder models like E5-base-v2 rely on standard self attention mechanisms that scale as $\mathcal{O}(L^2)$ with sequence length L [4]. This scaling factor is a bottleneck and it is the reason for the stricter chunking limits of 512 tokens.

State Space Models and linear-attention architectures scale as $\mathcal{O}(L)$, and would therefore allow for entire multi-page legal sections to be encoded without

risk of truncation. Combining the existing thermodynamic diagnostics ΔS_{ret} with that type of architecture would be an appropriate way to measure if removing chunking constraints would further sharpen the retrieval distribution.

On the cloud side, frontier models’ context windows are growing quickly but they still have a cost and latency when evaluating millions of tokens per query. The local approach may not have the same context window but it is privacy preserving and has no per-query token cost. Both have their place, but in a system like ours, a local private model is the best for sensitive legal documents.

Conclusions

This project has shown that statistical mechanics and generative AI can be used together to improve retrieval in local RAG for credit agreements. Corpus analysis showed that a purely statistical approach to create training supervision would be very challenging. The language is too repetitive and too semantically related for proximity or frequency statistics to reliably separate relevant and irrelevant chunks. However, this negative result was useful and inspired the change to generative supervision.

The first sub claim was that a local LLM could generate enough training supervision signal without the need for manual annotations, cloud API or data leaving the environment. This claim is supported by the use of Gemma3-12B which ran locally via Ollama which made 12,520 synthetic triplets from section-aware chunks on an M4 MacBook Air with no dedicated GPU. This result showed that LLMs can create useful training data locally and do not need commercial infrastructure.

The second sub claim was that domain specific fine-tuning will improve the retrieval accuracy compared to an off-the-shelf embedding model. This claim was also supported through the use of section-aware chunking, hybrid negative mining and MNRL. E5-base-v2 improved in all five of the held out folds with Recall@10 increasing by 25.2% from 0.637 to 0.797, MRR improving by 24.9% from 0.387 to 0.483 and mean rank falling from 45.96 to 14.77. Overall, these results conclusively show that fine-tuning has made the embedding space more domain aware, and this has substantially improved reliability of the local retriever.

The third sub claim was that a sharper retrieval distribution will lower the effective context window needed to answer a user’s query and this has been

supported by the Boltzmann implementation. At the MNRL working point of $\tau = 0.05$, the mean retrieval entropy dropped from 10.49 bits to 4.75 bits after fine-tuning with the top-five probability mass increasing from 0.013 to 0.589. Using these statistics also allowed us to demonstrate that fewer tokens now have to be passed into the LLM. Before fine-tuning, to achieve 80% Recall@k required $k \approx 35$ chunks for the base E5 model but only $k \approx 11$ chunks for the fine-tuned E5 model.

Overall, the answer to the research question is yes, but with limitations. Statistical mechanics alone struggled to create the training data, but it did allow us to explain why a purely statistical route failed. Statistical mechanics also provided the useful and necessary tools to measure the improvement in retrieval distributions after fine-tuning. The use of generative AI supplied the missing semantic supervision needed and along with contrastive fine-tuning, managed to reshape the embedding landscape. Together, this allowed more relevant chunks to occupy lower effective energy states. As a result, the project produced a more reliable local retriever, with a sharper, lower entropy retrieval distribution and a reduced reliance on large downstream context windows.

Acknowledgements

I would like to thank my project supervisor Sergey for their valuable time, experience, guidance and feedback throughout the project. I am also grateful to my industry partner for giving me the opportunity to learn something new and for all the guidance they provided over the year. I would also like to thank Loughborough University for providing the hardware needed for fine-tuning.

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A Statement of contribution

All experimental design, corpus statistics, chunking design, negative mining, fine-tuning, evaluation, thermodynamic interpretations and report writing was carried out by F227137. All code can be found on [GitHub](#)

Specific technical contributions include:

- Extracting and cleaning text from five PDF credit agreements using pdfplumber storing data as .jsonl
- Making a customised five stage section-aware chunking algorithm after paragraph chunks failure
- Generating 12,520 synthetic training triplets locally using Gemma3-12B via Ollama
- Writing a custom training pipeline using PyTorch DataLoaders and Sentence-Transformers to fine-tune E5-base-v2 and GTE-ModernBERT-base with MNRL on a single DigiLabs GPU
- Writing Python scripts to visualise the underlying statistics of the dataset
- Handover of fold-4 model weights to the industry partner, 84% Recall@10

Declaration of Generative AI Use: Generative AI tools were used in accordance to the university guidelines and played a supportive role:

- **Coding Assistance:** Throughout the project AI assisted with debugging Python and Pytorch code and ensuring visuals were neat. All of the logic was first researched by me and then attempted alone first. All AI suggested improvements were understood and fully tested before implementation.
- **Proofreading:** Generative AI was used for proofreading, fixing grammatical errors, and latex help. No AI tools were used for text writing or the interpretation/discussion of results.
- **Literature Search:** AI tools were used throughout the project to help with researching ideas for the project direction.

B Per-Fold Results

Fold	Model	#Triplets	R@1	R@5	R@10	MRR	Mean Rank	Median Rank	Mean Margin	S_{ret}	Top-5 Mass	Top-10 Mass
1	Base E5	2200	0.252	0.525	0.621	0.377	49.7	5	0.018	10.49	0.013	0.023
1	Fine-tuned E5	2200	0.344	0.684	0.798	0.491	14.9	3	0.183	4.79	0.587	0.687
2	Base E5	1744	0.274	0.523	0.642	0.393	59.4	5	0.018	10.50	0.012	0.022
2	Fine-tuned E5	1744	0.315	0.667	0.774	0.468	16.7	3	0.203	4.57	0.612	0.705
3	Base E5	3800	0.239	0.486	0.604	0.359	48.2	6	0.016	10.49	0.013	0.023
3	Fine-tuned E5	3800	0.318	0.640	0.776	0.467	17.5	3	0.156	5.15	0.540	0.641
4	Base E5	1608	0.311	0.588	0.703	0.437	31.3	4	0.023	10.49	0.013	0.023
4	Fine-tuned E5	1608	0.348	0.729	0.842	0.506	9.5	3	0.219	4.34	0.640	0.735
5	Base E5	3168	0.249	0.516	0.613	0.369	41.2	5	0.017	10.49	0.013	0.022
5	Fine-tuned E5	3168	0.330	0.682	0.793	0.485	15.3	3	0.172	4.93	0.565	0.666

Table 6: Per-fold E5 retrieval and thermodynamic diagnostic results with section-aware chunking. S_{ret} , Top-5 mass and the Top-10 mass are calculated at $\tau = 0.05$.

C Synthetic Query Generation Prompt

The following prompt was used with Gemma3-12B via Ollama to generate the synthetic training queries from each section-aware chunk. The prompt was refined using GPT-5.4 before being applied locally.

You are generating high-quality training data for a semantic retrieval system over credit agreements (legal financial documents).

Your goal is to create diverse, realistic search queries that professionals (e.g., lawyers, credit analysts, treasury teams, compliance officers) would

use to retrieve the paragraph below.

Section context: {section_heading}

Instructions:

Generate exactly 4 DISTINCT queries. Each query must follow a DIFFERENT style:

1. Precise factual query
Ask about a specific obligation, threshold, party, definition, trigger, or condition.
2. Paraphrased / indirect query
Rephrase the meaning without copying wording from the paragraph.
3. Keyword-style search
2-6 words, no question mark, like a real search input.
4. Analytical / scenario-based query
Ask how, why, when, or what happens under certain conditions.

Critical rules:

- Queries must be answerable primarily from this paragraph
- Do NOT reference section numbers, clause numbers, or headings explicitly
- Do NOT copy long phrases verbatim unless legally necessary
- Prefer semantic variation and synonyms over exact phrasing
- Avoid making all queries definition-style
- At least one query should focus on implication, usage, or consequence
- At least one query must use wording that differs from key terms
- Avoid vague or generic queries
- Ensure variation in length, structure, and intent
- Write as a human professional would naturally search

Output rules:

- Output ONLY the queries
- One query per line
- Numbered 1, 2, 3, 4, ...
- No explanations, no headings, no blank lines

Paragraph:

```
""  
{chunk_text}  
""
```

Queries:

D Additional Figures

Table 7: Baseline retrieval performance of seven potential embedding models evaluated on paragraph chunks before fine-tuning. E5-base-v2 was chosen as the primary model due to pipeline compatibility with the industry partner’s existing infrastructure. GTE-ModernBERT-base was retained as an auxiliary model due to its larger context window and promising results. All-mpnet-base-v2 is the industry partner’s current production model.

Model	Params	Dim	MRR	R@1	R@5	R@10	Mean Rank	Mean Margin	% Margin > 0
all-mpnet-base-v2	110M	768	0.403	29.20%	52.90%	62.70%	81.5	-0.0486	34.10%
bge-base-en-v1.5	109M	768	0.464	35.40%	59.00%	67.20%	73.7	-0.0187	40.70%
bge-base-financial-matryoshka	109M	768	0.38	28.70%	48.40%	55.00%	133.8	-0.0788	33.10%
e5-base-v2	109M	768	0.502	38.50%	64.10%	72.50%	42.8	-0.0046	44.40%
granite-embedding-english-r2	149M	768	0.48	36.20%	61.80%	71.30%	39.2	-0.0078	41.90%
gte-modernbert-base	149M	768	0.51	37.80%	66.70%	76.90%	25.8	-0.0089	44.40%
mxbai-embed-large-v1	335M	1024	0.5	38.60%	63.40%	72.10%	52.4	-0.0136	44.20%

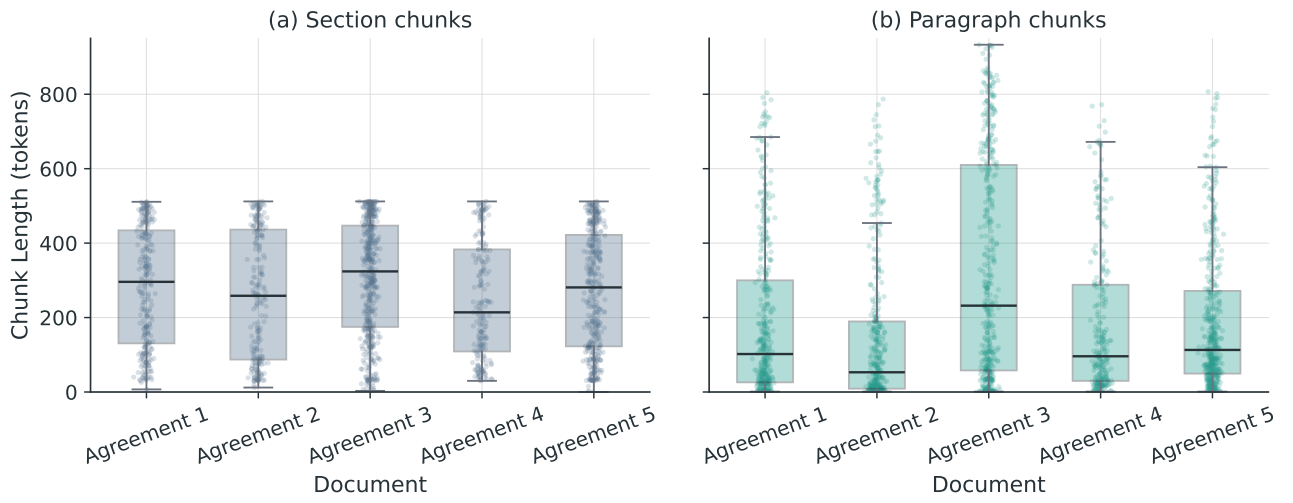


Figure 11: Chunk length distributions in tokens across the five credit agreements for section-aware chunking and paragraph chunking. Section chunks are bounded capped at 512 tokens by design. Paragraph chunks regularly exceed 512 tokens, with some reaching 800–900 tokens. This confirms that paragraph chunking caused substantial truncation when passed into the embedding model.

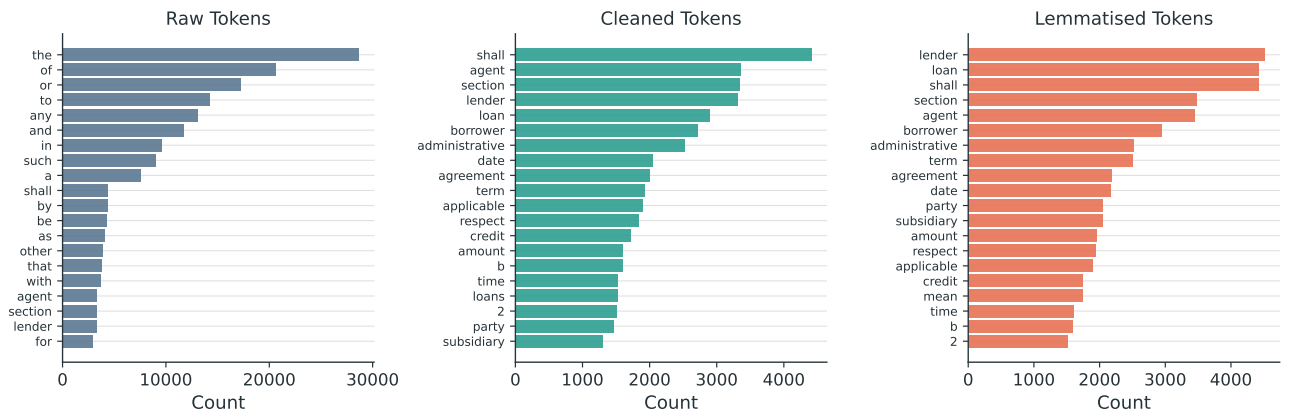


Figure 12: Top-20 token frequency comparison plot across three preprocessing stages: raw tokens (left), after stop-word removal (centre) and after lemmatisation (right). In the raw corpus, stop words like “the” and “of” dominate. After cleaning we are left with domain-specific legal terms including “lender”, “loan”, “agent” and “shall” emerge as the most frequent tokens, confirming the compressed mid-frequency legal vocabulary identified by Zipf’s law.

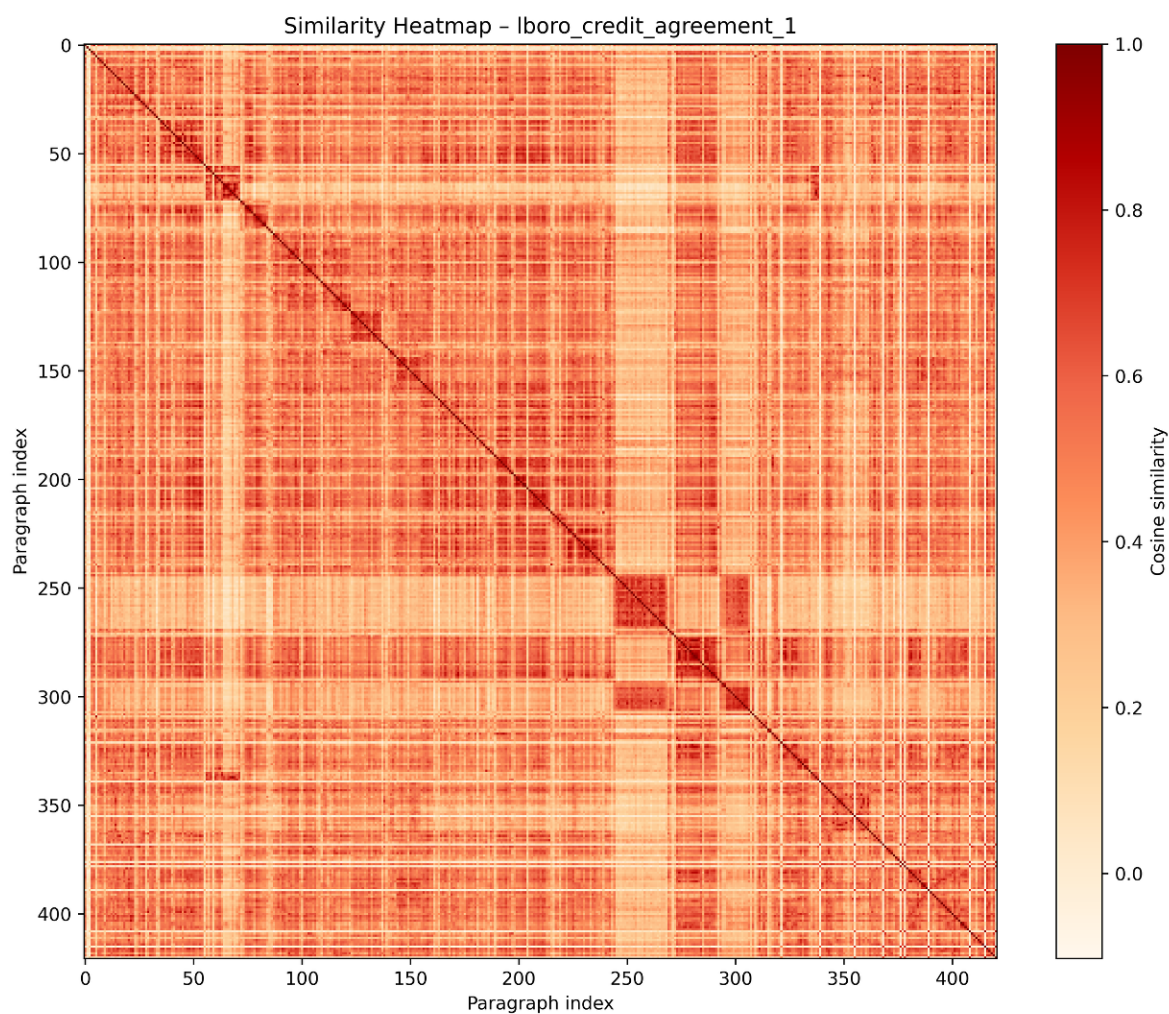


Figure 13: Cosine similarity heatmap for paragraph chunks from credit agreement 1 under the base E5 model. The dark diagonal line is a perfect chunk match and the blocks along it indicate regions of higher semantic relationships.